



Ecuatii LINIARE cu coef. CONSTANTİ

Forma generală: $x''(t) + a \cdot x'(t) + b \cdot x(t) = f(t)$
 $a, b \in \mathbb{R}$ (constante), $f \in C[a, b]$

$f(t) \neq 0 \rightarrow$ ec. liniară neomogenă cu coef. constante

Et1: x_0

Et2: x_p

Et3: $x = x_0 + x_p$

$= 0 \Rightarrow x'' + a \cdot x' + b \cdot x = 0$

ec. lin. omogenă cu coef. const.

Cazul omogen: $x''(t) + a \cdot x'(t) + b \cdot x(t) = 0$

Se caută o soluție de forma: $x(t) = e^{\lambda t}$

$$(e^{\lambda t})'' + a \cdot (e^{\lambda t})' + b \cdot e^{\lambda t} = 0$$

$$\lambda^2 e^{\lambda t} + a \lambda e^{\lambda t} + b e^{\lambda t} = 0$$

$$(\lambda^2 + a \lambda + b) e^{\lambda t} = 0 \quad | : e^{\lambda t} \neq 0$$

$$\lambda^2 + a \lambda + b = 0 \quad \text{ecuație caracteristică}$$

$\Rightarrow \lambda_1, \lambda_2 = \text{solutiile ec. caracteristice.}$

Discuție : ($\Delta = a^2 - 4b$ - discrimin. ec. caract)

a) $\Delta > 0 \Rightarrow \lambda_1 \neq \lambda_2 \in \mathbb{R} \Rightarrow \begin{cases} x_1 = e^{\lambda_1 t} \\ x_2 = e^{\lambda_2 t} \end{cases}$

Teorema $\{x_1, x_2\}$ - formația s. fundam. de soluții
ptr. ec. dif omogenă cu coef const

b) $\Delta = 0 \Rightarrow \lambda_1 = \lambda_2 \in \mathbb{R} \Rightarrow \begin{cases} x_1 = e^{\lambda_1 t} \\ x_2 = t e^{\lambda_1 t} \end{cases}$

Teorema $\{x_1, x_2\}$ - s. fundam. sl.

c) $\Delta < 0 \Rightarrow \lambda_{1,2} = \alpha \pm i\beta \in \mathbb{C} \setminus \mathbb{R}$
 $\Rightarrow \begin{cases} x_1 = e^{\alpha t} \cos \beta t \\ x_2 = e^{\alpha t} \sin \beta t \end{cases}$

Teorema $\{x_1, x_2\}$ - s. fundam. sl.

$\hookrightarrow$ de asemenea
 $\bullet x_1, x_2 = \text{sol.}$
 $\bullet x_1, x_2 = \text{lin. indep. } (\forall (t; x_1, x_2) \neq 0)$

Cazul neomogen $x''(t) + a \cdot x'(t) + b x(t) = f(t)$

$$a, b \in \mathbb{R}, f \in C[a, b]$$

Et1: se rezolvă ec. omogenă :

$$X_{\bullet} = c_1 \cdot x_1 + c_2 \cdot x_2$$

unde $\{x_1, x_2\} = \text{s.f. sol.}$

Et2: se caută o sol. particulară x_p :

x_p : met. Lagrange
(metoda variației constantelor)
(vezi curs anterior)

! valabilă ptr orice tip de $f(t)$

metoda coeficienților nedeterminați

! valabilă pentru anumite forme ale lui $f(t)$

Metoda coeficientilor nedeterminati

① $f(t) = P_m(t)$ - polinom de grad m

$$\Rightarrow x_p = \begin{cases} Q_m(t) & , \text{dacă } h_1, h_2 \neq 0 \\ t \cdot Q_m(t) & , \text{dacă } h_1 = 0 \\ t^2 \cdot Q_m(t) & , \text{dacă } h_1 = h_2 = 0 \end{cases}$$

$h_1, h_2 = \text{rad ee. caract} : r^2 + ar + b = 0$

② $f(t) = e^{\alpha t} \cdot P_m(t)$

$$\Rightarrow x_p = \begin{cases} e^{\alpha t} \cdot Q_m(t) & , \text{dacă } h_1, h_2 \neq \alpha \\ t \cdot e^{\alpha t} \cdot Q_m(t) & , \text{dacă } h_1 = \alpha \\ t^2 \cdot e^{\alpha t} \cdot Q_m(t) & , \text{dacă } h_1 = h_2 = \alpha \end{cases}$$

③ $f(t) = e^{\alpha t} \cdot P_m(t) \cdot \cos pt$

sau

$$f(t) = e^{\alpha t} \cdot P_m(t) \cdot \sin pt$$

$$\Rightarrow x_p = e^{\alpha t} [Q_m(t) \cos t + R_m(t) \sin t] , \text{ dacă } \lambda_{1,2} \neq \alpha \pm i\beta$$

$$\left\{ \begin{array}{l} x_p = t e^{\alpha t} [Q_m(t) \cos t + R_m(t) \sin t] \\ \text{dacă } \lambda_{1,2} = \alpha \pm i\beta \end{array} \right.$$

Principiul superpoziției

Dacă termenul necomogen al ec. necomogene
 cu coef. const. se poate "desface" în
 sumă de funcții cu caracteristici descrise
 în cazurile A, B, C, atunci r_l ml.
 particulară se va calcula distinct
 pentru fiecare din cazurile menționate.

$$f(t) = f_1(t) + f_2(t)$$

$$\begin{array}{cc} \downarrow & \downarrow \\ x_{p1}(t) & x_{p2}(t) \end{array}$$

$$x_p = x_{p1}(t) + x_{p2}(t)$$

$$x'' + a \cdot x' + b \cdot x = f_1(t) + f_2(t)$$

Et1: $x'' + ax' + bx = 0 \Rightarrow x_0 = \dots$

Et2:

(2.1.) $x'' + ax' + bx = f_1(t) \Rightarrow x_{p1} = \dots$

(2.2.) $x'' + ax' + bx = f_2(t) \Rightarrow x_{p2} = \dots$

$$x_p = x_{p1} + x_{p2}$$

Et3: $x = x_0 + x_p.$

ex1: $x'' - x = 0.$

ec. caract: $\lambda^2 - 1 = 0 \Rightarrow \lambda_{1,2} = \pm 1$

$$\Rightarrow x_1 = e^t, x_2 = e^{-t}$$

$$\Rightarrow \boxed{x = c_1 \cdot e^t + c_2 \cdot e^{-t}}, c_1, c_2 \in \mathbb{R}$$

ex2: $x'' + 2x' + x = 0.$

ec caract: $\lambda^2 + 2\lambda + 1 = 0 \Rightarrow \lambda_1 = \lambda_2 = -1$

$$\Rightarrow x_1 = e^{-t}, x_2 = t \cdot e^{-t}$$

$$\Rightarrow \boxed{x = c_1 e^{-t} + c_2 t e^{-t}}, c_1, c_2 \in \mathbb{R}$$

ex: $x'' + x = 0$

characteristic: $\lambda^2 + 1 = 0 \Rightarrow \lambda_{1,2} = \pm i = 0 \pm 1 \cdot i$

$x_1 = e^{0t} \cdot \cos(1 \cdot t)$, $x_2 = e^{0t} \sin(1 \cdot t)$

$\Rightarrow \boxed{x = c_1 \cdot \cos t + c_2 \sin t}$, $c_1, c_2 \in \mathbb{R}$

ex: $x'' - x = \underbrace{3e^t}_{f_1(t)} + \underbrace{5t + 6}_{f_2(t)} + \underbrace{\cos t}_{f_3(t)}$

Et1: $x'' - x = 0 \Rightarrow x_0 = c_1 \cdot e^t + c_2 e^{-t}, c_{1,2} \in \mathbb{R}$
 $(\lambda_{1,2} = \pm 1)$

Et2:

(2.1) $f_1(t) = 3e^t$
 $\hookrightarrow \alpha = 1, m = 0$

$\Rightarrow \boxed{x'' - x = 3e^t}$

$x_{p1} = Q_0(t) \cdot e^t \cdot t \rightarrow \alpha = r_1 = 1$

$x_{p1} = a \cdot e^t \cdot t$ sol $x_{p1}'' - x_{p1} = 3e^t$

$x_{p1}' = a e^t + a t e^t$; $x_{p1}'' = a e^t + a e^t + a t e^t$

$\Rightarrow 2a e^t + \cancel{a t e^t} - \cancel{a t e^t} = 3e^t \Rightarrow a = \frac{3}{2}$

$$\Rightarrow x_{p1} = \frac{3}{2} t e^t$$

2.2.

$$f_2 = \underbrace{5t + 6}_{P_1(t)}$$

$$\Rightarrow x'' - x = 5t + 6$$

$$\Rightarrow x_{p2} = Q_1(t) = at + b$$

$$x_{p2}'' - x_{p2} = 5t + 6$$

$$x_{p2}' = a$$

$$x_{p2}'' = 0$$

$$\stackrel{ec}{\Rightarrow} 0 - at - b = 5t + 6 \quad \Rightarrow \begin{aligned} a &= -5 \\ b &= -6 \end{aligned}$$

$$\Rightarrow x_{p2} = -5t - 6$$

2.3

$$f_3 = \cos t = P_0(t) \cdot e^{0 \cdot t} \cdot \cos(1 \cdot t)$$

$$m=0, \alpha=0, p=1$$

$$\alpha + ip = i \neq \lambda_{1,2}$$

$$x'' - x = \cos t$$

$$x_{p3} = \int_{-1}^1 Q_0(t) \cdot \cos t + R_0(t) \cdot \sin t$$

$$x_{p3} = a \cdot \cos t + b \cdot \sin t$$

$$x_{p3}'' - x_{p3} = \cos t$$

$$x_{p3}' = -a \sin t + b \cos t$$

$$x_{p3}'' = -a \cos t - b \sin t$$

$$\stackrel{ec}{\Rightarrow} -a \cos t - b \sin t - a \cos t - b \sin t = \cos t$$

$$-2a \cos t - 2b \sin t = \cos t \Rightarrow a = -\frac{1}{2}$$

$$b = 0.$$

$$\Rightarrow x_{p3} = -\frac{1}{2} \cos t.$$

$$\Rightarrow x_p = x_{p1} + x_{p2} + x_{p3}$$

$$x_p = \frac{3}{2} t e^t - 5t - 6 - \frac{1}{2} \cos t$$

It3:

$$x = c_1 e^t + c_2 e^{-t} + \frac{3}{2} t e^t - 5t - 6 - \frac{1}{2} \cos t$$

$$c_1, c_2 \in \mathbb{R}$$

Sisteme de ecuații diferențiale

Sisteme de ecuații diferențiale liniare de ordinul 2.

Forma generală:

$$(1) \begin{cases} x_1'(t) = a_{11}(t) \cdot x_1(t) + a_{12}(t) \cdot x_2(t) + b_1(t) \\ x_2'(t) = a_{21}(t) \cdot x_1(t) + a_{22}(t) \cdot x_2(t) + b_2(t) \end{cases}$$

$$\begin{cases} a_{ij} \in C([a, b]; \mathbb{R}) , b_i \in C([a, b]; \mathbb{R}) , i, j = 1, 2 \\ \begin{cases} x_1 = x_1(t) \\ x_2 = x_2(t) \end{cases} \end{cases}$$

memorizatele sistemului

$$[a, b] \subseteq \mathbb{R}$$

$$(1) \boxed{X' = A \cdot X + B} \text{ - sist. lin. neomogen}$$
$$A = \begin{pmatrix} a_{11}(t) & a_{12}(t) \\ a_{21}(t) & a_{22}(t) \end{pmatrix} ; B = \begin{pmatrix} b_1(t) \\ b_2(t) \end{pmatrix} ; X(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$$
$$(2) \boxed{X' = AX} \text{ - sist. lin. omogen}$$

Fie operatorul $L: C^1([a, b]; \mathbb{R}^2) \rightarrow C([a, b]; \mathbb{R}^2)$

definit prin: $X \rightarrow \boxed{LX := X' - AX}$

Dacă $\nearrow LX = 0 \quad (\Leftrightarrow X' - AX = 0 \text{ - caz. omogen})$
 $\searrow LX = B \quad (\Leftrightarrow X' - AX = B \text{ - caz. neomogen})$

Notăm:

$$S_0 := \{X \in C^1([a, b]; \mathbb{R}^2), LX = 0\} = \ker L$$

$\hookrightarrow$ mulțimea sol. sist. lin. omogen

$$S := \{X \in C^1([a, b]; \mathbb{R}^2), LX = B\}$$

$\hookrightarrow$ mulțimea sol. sist. lin. neomogen

Teorema: ($\nexists$! a problemei Cauchy)

Problema Cauchy (problema cu valori inițiale)

$$\begin{cases} LX = B \\ X(t_0) = \alpha \end{cases}, \quad t_0 \in [a, b], \alpha := \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

sunt soluții unice: $X(\cdot, t_0, B, \alpha)$.

$X(\cdot, t_0, 0, \alpha)$ - sol p/b Cauchy

$$\Downarrow \\ B=0$$

ec.:

$$LX=0$$

$$\Updownarrow \\ A'X - X = 0$$

cond: $X(t_0) = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \in \alpha$

$X(\cdot, \underset{t_0=0}{0}, 0, \begin{pmatrix} 1 \\ 0 \end{pmatrix})$ - s.f. $LX=0$

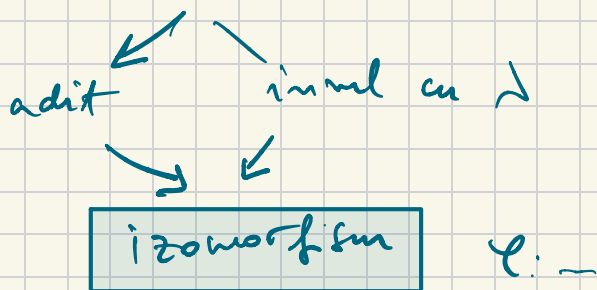
$$\begin{cases} X(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{cases}$$

I. Sisteme liniare omogene (Cazul omogen)

$$\begin{cases} x_1'(t) = a_{11}(t) x_1(t) + a_{12}(t) x_2(t) \\ x_2'(t) = a_{21}(t) x_1(t) + a_{22}(t) x_2(t) \end{cases}$$

(P1) Multimea soluțiilor sist. omogen, S_0 ,
formează subspațiul liniar bidimensional.

demonstrăm: $\rightarrow S_0 = \text{sp. liniar} \subseteq C^1[a, b]$



$\rightarrow$ dim $S_0 = 2$, adică $\{X^1, X^2\}$ - bază S_0
adică $\left\{ \begin{array}{l} \forall X \in S_0 \exists c_1, c_2 \in \mathbb{R} \text{ a.c.} \\ X = c_1 X^1 + c_2 X^2 \\ X^1, X^2 = \text{sol} \in S_0 \end{array} \right.$

Obs: $\{X^1, X^2\}$ -s. f. soluții dacă

$X^1 = \text{sol sist omog}$ $\&$ $X^2 = \text{sol sist omog}$ $X^1, X^2 = \text{l. indep.}$
 $(W(t; X^1, X^2) \neq 0)$

Obs: $U(t) = \begin{pmatrix} X^1 & X^2 \end{pmatrix} = \text{matrice fundam}$
de soluții

$X^0 = U(t) \cdot \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}, c_1, c_2 \in \mathbb{R}$

forma sol. sist omogen.

ex: Fie sistemul:

$$\begin{cases} x_1'(t) = \left(1 + \frac{1}{t}\right) x_1(t) - \frac{1}{t} x_2(t) \\ x_2'(t) = \frac{t-2}{t-1} x_1(t) + \frac{1}{t-1} x_2(t) \end{cases}$$

Arătați că funcțiile X^1, X^2 - s. f. sol.

$$X^1 = \begin{pmatrix} t \\ t^2 \end{pmatrix} ; X^2 = \begin{pmatrix} e^t \\ e^t \end{pmatrix}$$

Soluție:

$$\rightarrow X^1 = \text{sol. sist?} \quad \begin{cases} x_1' = t \\ x_2' = t^2 \end{cases} \text{ sol. sist.}$$

$$\begin{cases} t' = \left(1 + \frac{1}{t}\right) \cdot t - \frac{1}{t} \cdot t^2 \quad \checkmark \\ (t^2)' = \frac{t-2}{t-1} \cdot t + \frac{1}{t-1} \cdot t^2 \quad \checkmark \end{cases}$$

$$\rightarrow X^2 = \text{sol. sist?}$$

Teorema

$$\rightarrow \{X^1, X^2\} \text{ lin indep} ; W(t; X^1, X^2) = \begin{vmatrix} t & e^t \\ t^2 & e^t \end{vmatrix} \neq 0$$

$$\Rightarrow \{X^1, X^2\} = \text{a. f. sol.}$$

$$U(t) = (X^1 X^2) = \begin{pmatrix} t & e^t \\ t^2 & e^t \end{pmatrix} = \text{m.f. sol.}$$

$$X = U(t) \cdot \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

$$X = \begin{pmatrix} t & e^t \\ t^2 & e^t \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \quad \text{sol. int.}$$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} c_1 t + c_2 e^t \\ c_1 t^2 + c_2 e^t \end{pmatrix}$$

$$\begin{cases} x_1 = c_1 t + c_2 e^t \\ x_2 = c_1 t^2 + c_2 e^t \end{cases} \quad \text{sol. int.}, \quad c_1, c_2 \in \mathbb{R}$$

II Sisteme liniare neomogene (Cazul neomogen)

$$X' = AX + B$$

Teorema

$$S = S_0 + \{X^P\}$$

sol. generală
sist. neomogen

sol. generală
a sistemului omogen

sol. partic.
sist.
neomogen

X^P = soluție particulară (sist neomogen)
→ Metoda variației constantelor

$$X^P(t) = U(t) \cdot \begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix}, \quad U(t) = (X^1 \ X^2)$$

$$(X^P(t))' - A(t) \cdot X^P(t) = B(t)$$

$$\left[U(t) \begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix} \right]' - A \cdot \left[U \cdot \begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix} \right] = B$$

$$U'(t) \cdot \begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix} + U(t) \cdot \begin{pmatrix} c_1'(t) \\ c_2'(t) \end{pmatrix} - A \cdot U(t) \cdot \begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix} = B(t)$$

$$\underbrace{[U'(t) - A \cdot U(t)]}_{=0} \cdot \begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix} + U(t) \cdot \begin{pmatrix} c_1'(t) \\ c_2'(t) \end{pmatrix} = B(t)$$

choose $U(t) = \text{m.f. sol.}$

independent, first ones

$$\Rightarrow U(t) \cdot \begin{pmatrix} c_1'(t) \\ c_2'(t) \end{pmatrix} = B(t) \Rightarrow \begin{pmatrix} c_1'(t) \\ c_2'(t) \end{pmatrix} = U^{-1} \cdot B$$

$$\Rightarrow \begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix} = \int U(t)^{-1} \cdot B(t) dt$$

$$\Rightarrow X^p = U(t) \cdot \int U(t)^{-1} \cdot B(t) dt$$

ex: Fie sistemul:

$$\begin{cases} x_1'(t) = \left(1 + \frac{1}{t}\right) x_1(t) - \frac{1}{t} x_2(t) + 1 \\ x_2'(t) = \frac{t-2}{t-1} x_1(t) + \frac{1}{t-1} x_2(t) + t \end{cases}$$

Calculați sol. generală

indicată

X^e = se preia de la ex
anterior

$$\cancel{X}^P = \begin{pmatrix} t & e^t \\ t^2 & e^t \end{pmatrix} \begin{pmatrix} c_1(t) \\ c_2(t) \end{pmatrix}$$

$$X = X^o + X^P$$

...

$$\begin{cases} x_1 = \\ x_2 = \end{cases}$$

